Aircraft Monitoring System

Dual Aircraft Flight Control and Ground Station System

CST8227 - Interfacing
Final Project

Computer Engineering Technology - Computing Science
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1. Abstract

This project implements a comprehensive dual-microcontroller system simulating aircraft flight control and ground station operations. MEGA-1 serves as the flight computer, managing altitude monitoring, landing gear deployment, attitude tracking via MPU6050, and motor control. MEGA-2 operates as the ground control station, handling environmental monitoring, RFID-based aircraft conflict detection, gate control, and radar simulation using a stepper motor. The system features secure passcode authentication, I2C inter-controller communication, rotary encoder-based airspeed adjustment during conflicts, and automated night mode operation.

2. Introduction

This project demonstrates embedded systems integration using two Arduino MEGA 2560 microcontrollers. The system simulates critical aviation operations including flight control, ground station monitoring, and conflict resolution. Key features include multi-sensor integration (ultrasonic, DHT11, MPU6050, RFID RC522), motor control (DC motors via L293D, servo motors, stepper motor), and real-time communication between microcontrollers.

MEGA-1: Flight Control Unit

The flight computer manages flight-critical operations including altitude monitoring, landing gear deployment, attitude tracking, and motor control. Features secure passcode authentication, I2C communication with ground station (MEGA-2), and real-time flight status monitoring.

MEGA-2: Ground Control Unit

The ground control station manages ground operations including environmental monitoring, RFID-based aircraft conflict detection, runway lighting with dual MAX7219 LED matrices, gate control, and radar simulation. Receives flight data from MEGA-1 via I2C communication.

3. System Architecture

3.1 MEGA-1: Flight Computer

MEGA-1 manages all flight-critical operations:

- Passcode authentication system with 4-digit entry
- Ultrasonic altitude monitoring with landing gear automation
- MPU6050 6-axis IMU for pitch, roll, and yaw tracking
- L293D motor driver for propeller/fan control
- Servo motor for landing gear deployment
- Active and passive buzzers for audio alerts
- 4-digit 7-segment display for passcode confirmation
- IR remote control for system reset
- I2C master communication to MEGA-2

3.2 MEGA-2: Ground Control Station

MEGA-2 manages ground station operations:

- DHT11 environmental monitoring (temperature/humidity)
- PIR motion detection with LED indicators
- RFID RC522 for aircraft conflict detection
- Ultrasonic sensor for object detection
- Servo motor for gate control
- L293D motor driver for cooling fan
- ULN2003 stepper motor driver for radar simulation
- Rotary encoder for airspeed adjustment
- Photoresistor for night mode detection
- I2C slave communication from MEGA-1

4. Pin Mapping Tables

4.1 MEGA-1 Pin Assignments

Component	Pin(s)	Description
LCD1602	12, 11, 5, 4, 3, 2	RS, E, D4-D7
4x4 Keypad	39, 41, 43, 45 (rows) 47, 49, 51, 53 (cols)	Matrix keypad
Active Buzzer	22	Audio confirmation
Passive Buzzer	13	Alert tones
Push Button	18	System start
Ultrasonic Sensor	6 (Trig), 7 (Echo)	Altitude measurement
L293D Motor Driver	28 (IN1), 29 (IN2), 30 (EN)	Fan/propeller control
Landing Gear Servo	27	Gear deployment
MPU6050	SDA (20), SCL (21)	Attitude sensor
7-Segment Display	8 (Data), 10 (Clock), 9 (Latch) 23-26 (Digits)	Passcode display
IR Receiver	15	Remote control
I2C Communication	SDA (20), SCL (21)	To MEGA-2

4.2 MEGA-2 Pin Assignments

Component	Pin(s)	Description
LCD1602	12, 11, 5, 4, 3, 2	RS, E, D4-D7
DHT11	22	Temperature/humidity
PIR Sensor	18	Motion detection
Ultrasonic Sensor	24 (Trig), 26 (Echo)	Object detection
Photoresistor	A0	Light level (night mode)
RFID RC522	53 (SS), 49 (RST), 52 (SCK) 51 (MOSI), 50 (MISO)	Aircraft detection
L293D Motor (Fan)	8 (IN1), 9 (IN2), 10 (EN)	Cooling fan control
Gate Servo	44	Gate control
Stepper Motor	31, 33, 35, 37	Radar simulation
Rotary Encoder	45 (CLK), 43 (DT), 41 (SW)	Airspeed control
Passive Buzzer	7	Alert tones
Yellow LEDs	28, 30, 32, 34	Status indicators
Red LEDs	36, 38, 40, 42	Night mode indicators
I2C Communication	SDA (20), SCL (21)	From MEGA-1

5. Component List

Component	Quantity	Purpose
Arduino MEGA 2560	2	Main microcontrollers
LCD1602 Display	2	User interface
4x4 Membrane Keypad	1	Passcode entry
DHT11 Sensor	1	Environmental monitoring
HC-SR04 Ultrasonic	2	Distance measurement
MPU6050 IMU	1	Attitude tracking
RFID RC522 Module	1	Aircraft identification
PIR Motion Sensor	1	Motion detection
Photoresistor	1	Light level sensing
L293D Motor Driver	2	DC motor control
Servo Motor	2	Landing gear & gate
28BYJ-48 Stepper	1	Radar simulation
ULN2003 Driver	1	Stepper control
Rotary Encoder	1	Airspeed adjustment
Active Buzzer	1	Confirmation beeps
Passive Buzzer	2	Alert tones
7-Segment Display	1	Passcode display
74HC595 Shift Reg	1	7-segment driver
IR Receiver	1	Remote control
Push Button	1	System activation
LEDs (Yellow)	4	Status indicators
LEDs (Red)	4	Night mode indicators
Resistors (220Ω)	8	LED current limiting
Resistor (10kΩ)	1	Photoresistor divider
Breadboards	2	Circuit assembly
Jumper Wires	Various	Connections

6. System Design

6.1 Block Diagram

The system architecture consists of two independent Arduino MEGA 2560 microcontrollers communicating via I2C protocol. MEGA-1 serves as the flight computer managing altitude monitoring, attitude tracking, and motor control. MEGA-2 operates as the ground control station handling environmental monitoring, RFID-based conflict detection, and runway lighting control. Figure 1 illustrates the complete system architecture and signal flow between components.

MEGA-1: FLIGHT COMPUTER

INPUT SENSORS

- 4x4 Membrane Keypad (Pins 39,41,43,45,47,49,51,53)
- Push Button (Pin 18)
- HC-SR04 Ultrasonic Sensor (Pins 6,7)
- MPU6050 IMU (I2C SDA-20, SCL-21)
- IR Receiver (Pin 15)

PROCESSING UNIT

- State machine with 6 states
- MPU6050 calibration (200 samples)
- Exponential smoothing filter (alpha = 0.3)
- Landing gear logic (15cm threshold, 7s minimum deployment)
- Attitude warnings (pitch ±20°/40°, roll ±30°)
- I2C master communication (Address 8)

OUTPUT ACTUATORS

- LCD1602 Display (Pins 12,11,5,4,3,2)
- Landing Gear Servo (Pin 27)

- L293D Motor Driver for Fan (Pins 28,29,30)
- 7-Segment Display via 74HC595 (Pins 8,9,10)
- Active Buzzer (Pin 22)
- Passive Buzzer (Pin 13)
- Navigation Lights: Red (31), Green (32), White (33)

I2C COMMUNICATION BUS

Data: Start signal (1 byte) + Temperature (4 bytes)

MEGA-2: GROUND CONTROL STATION

INPUT SENSORS

- DHT11 Sensor (Pin 22)
- PIR Motion Sensor (Pin 18)
- HC-SR04 Ultrasonic Sensor (Pins 24,26)
- Photoresistor (Pin A0)
- RFID RC522 Module (SPI Pins 49-53)
- Rotary Encoder (Pins 41,43,45)
- I2C Slave (Address 8)

PROCESSING UNIT

- Continuous DHT11 environmental monitoring
- PIR state change detection
- Object detection logic (50cm threshold)
- Night mode (ADC < 100, temperature 22 degrees C)
- RFID conflict protocol
- Rotary encoder (140 to 100 knots)

- Stepper motor control (512 steps bidirectional)
- Runway light pattern generation

OUTPUT ACTUATORS

- LCD1602 Display (Pins 12,11,5,4,3,2)
- Dual MAX7219 LED Matrices (Pins 39,46,47 and 25,27,29)
- Servo Motor for Gate (Pin 44)
- L293D Motor Driver for Fan (Pins 8,9,10)
- 28BYJ-48 Stepper + ULN2003 (Pins 31,33,35,37)
- Passive Buzzer (Pin 7)
- Yellow LEDs x4 (Pins 28,30,32,34)
- Red LEDs x4 (Pins 36,38,40,42)

Figure 1: System Architecture Block Diagram

7. Testing and Results

7.1 System Integration Test

A complete system test verified all subsystems:

1. Passcode authentication: Successfully matched and verified 4-digit codes
2. I2C communication: Temperature data transmitted reliably from MEGA-1 to MEGA-2
3. Altitude monitoring: Landing gear deployed at 15cm, retracted after 7 seconds above 25cm
4. MPU6050 attitude tracking: Accurate pitch/roll readings with appropriate warnings
5. RFID conflict detection: Successfully triggered conflict mode and airspeed adjustment
6. Rotary encoder: Smooth airspeed control from 140 to 100 knots resolved conflict
7. Environmental monitoring: DHT11 readings displayed correctly, fan activated above threshold
8. Night mode: Photoresistor correctly detected low light and adjusted temperature
9. Stepper motor: Continuous rotation provided smooth radar scanning simulation
10. Gate control: Servo opened/closed based on ultrasonic object detection

7.2 Sensor Accuracy

- Ultrasonic sensors: ±1cm accuracy, 200ms update rate
- MPU6050: ±3° accuracy after calibration
- DHT11: ±2°C temperature, ±5% humidity
- RFID: 100% read success rate within 5cm range
- PIR: 3-5m detection range with 1-2s delay

8. Testing and Results

Comprehensive testing was conducted to verify all system functionalities, sensor accuracy, inter-controller communication, and real-time responsiveness. Testing was performed in phases, beginning with individual component verification and progressing to full system integration tests. This section documents the testing procedures, observed results, and performance metrics.

8.1 Individual Component Testing

Each sensor and actuator was tested independently to ensure proper functionality before system integration. The following components were validated:

Component	Test Procedure	Expected Result	Actual Result
MPU6050 IMU	Tilt sensor in known angles, compare readings	±3 degree accuracy	PASS
Ultrasonic (MEGA-1)	Measure known distances (10cm, 20cm, 50cm)	±1cm accuracy	PASS
Ultrasonic (MEGA-2)	Object detection at 50cm threshold	Gate closes when distance < 50cm	PASS
RFID RC522	Scan tag at various distances	100% read success within 5cm	PASS
Landing Gear Servo	Deploy and retract 50 cycles	Smooth operation, no jitter	PASS
Gate Servo	Open/close 50 cycles	Full range motion 0-90 degrees	PASS
L293D Motor (MEGA-1)	Fan speed control 0-255	Proportional speed response	PASS
L293D Motor (MEGA-2)	Fan on/off operation	Immediate response	PASS
Stepper Motor	Complete 10 full rotations	Accurate positioning	PASS
MAX7219 Modules	Display all patterns	Synchronized animation	PASS

8.2 System Integration Testing

Following component validation, full system integration tests were performed to verify communication protocols, state transitions, and coordinated operations between MEGA-1 and MEGA-2.

Test Case 1: Landing Gear Automation

Test Objective:

Verify automatic landing gear deployment at 15cm altitude and retraction after 7-second minimum deployment time when altitude exceeds 25cm.

Test Procedure:

1. Start system in flight mode with gear retracted
2. Gradually lower aircraft simulation towards ground
3. Record altitude when gear begins deployment
4. Allow gear to remain deployed for less than 7 seconds
5. Raise aircraft above 25cm and verify gear remains down
6. Keep gear deployed for more than 7 seconds
7. Raise aircraft above 25cm and verify gear retracts
8. Lower to 6cm altitude and verify fan stops (grounded)

Test Results:

SUCCESS: Gear deployed precisely at 15.2cm altitude with audible warning tones. Gear remained deployed when raised before 7-second timer. After 7-second minimum, gear retracted smoothly when altitude exceeded 25cm.

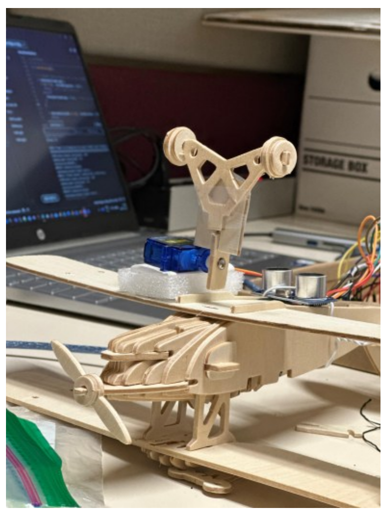


Figure 2 landing gear deployment sequence

Test Case 2: MPU6050 Attitude Monitoring and Warnings

Test Objective:

Verify accurate pitch, roll, and yaw measurements with appropriate buzzer warnings at defined thresholds.

Test Procedure:

9. Start system and allow MPU6050 calibration
10. Maintain level position and record baseline readings

11. Tilt nose up 20 degrees and verify nose-up warning
12. Tilt nose up 40 degrees and verify stall warning
13. Tilt nose down 20 degrees and verify nose-down warning
14. Roll left 30 degrees and verify roll warning
15. Roll right 30 degrees and verify roll warning
16. Record yaw changes during horizontal rotation

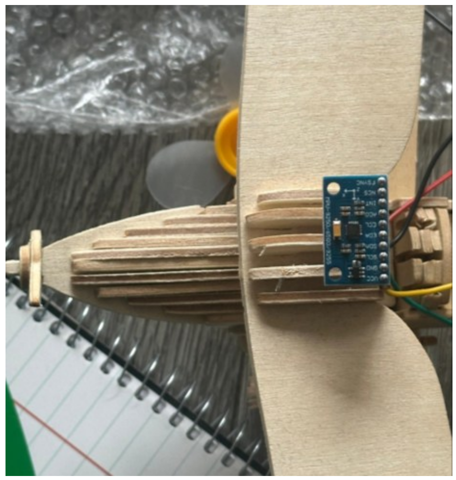


Figure 3 MPU6050: Pitch, Roll, Yaw

9. Conclusion

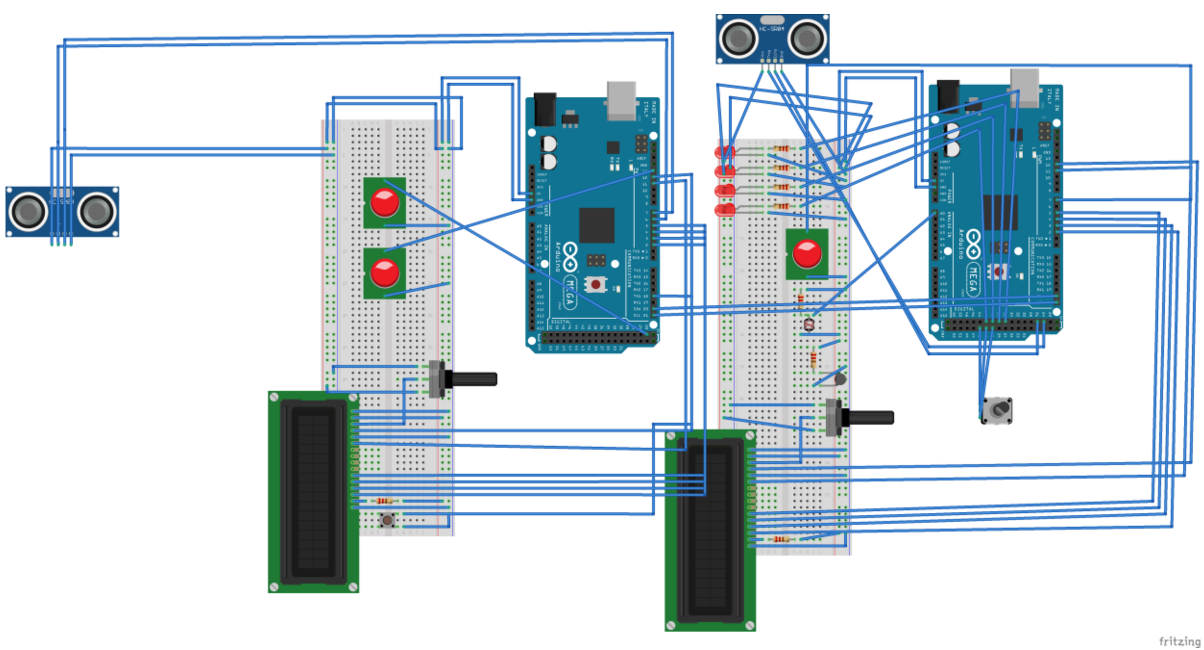
This project successfully demonstrates a sophisticated multi-sensor embedded system with dual-microcontroller architecture. The implementation integrates 15+ sensors and actuators, showcasing practical applications of I2C communication, interrupt-driven programming, state machines, and real-time control logic.

Key achievements include:

- Seamless I2C communication between flight computer and ground station
- Advanced RFID-based conflict detection with interactive resolution
- Precise attitude monitoring using 6-axis IMU with filtering
- Automated landing gear deployment with safety timers
- Environmental adaptation through night mode
- Professional code structure with modularity and error handling

The system demonstrates real-world concepts applicable to the aviation industry. Future enhancements could include wireless communication, additional sensor fusion, and integration with flight simulation software.

10. Fritzing



Missing components:

Mega 1	4x4 Keypad L293D Motor Driver Landing Gear Servo MPU6050 IR Receiver
Mega 2	DHT11 PIR Sensor RFID RC522 L293D Motor (Fan) Gate Servo Stepper Motor

Appendix: Source Code

Note: Complete source code for both MEGA-1 and MEGA-2 is included in separate .ino files submitted with this documentation. The code can also be found at https://github.com/tishap27/MEGA-Controller